Anurag Mishra

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EDUCATION

Master of Science, Artificial Intelligence 2023 – May 2025
Rochester Institute of Technology (RIT), Rochester, NY

Bachelor of Technology, Computer Science (Minor: Artificial Intelligence) 2019 – 2023
Sikkim Manipal Institute of Technology, India

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, SQL | **Frameworks:** PyTorch, JAX, TensorFlow, Transformers | **LLM Training:** SFT, LoRA, distributed training (DDP/FSDP), model evaluation
Performance: CUDA kernel development, multi-GPU optimization, workload profiling, benchmarking | **Systems:** Linux, Docker, HPC clusters, Git, CI/CD

PROFESSIONAL EXPERIENCE

AI Engineer Aug 2025 – Present
Eaton Ventures (Rochester Appliances), Rochester, NY

- Built production AI platform components for multimodal assistants serving 10K+ daily requests with 99.9% uptime
- Developed workload analysis tooling for model latency, throughput, and failure patterns; reduced end-to-end latency by 35%
- Implemented automated evaluation and regression workflows for model and system performance, reducing release cycle time by 60%
- Partnered with engineering and product teams to prioritize bottlenecks and ship reliability improvements into production

ML Research Assistant, DeFake Project (Multimodal Training) Oct 2024 – Aug 2025
Rochester Institute of Technology

- Trained multimodal CNN + transformer workloads on 18,000+ videos for lip-sync manipulation detection, improving F1 by 12%
- Implemented custom C++/CUDA kernels for forward/backward and cross-modal alignment paths, improving training efficiency by 40%
- Optimized distributed training with PyTorch DDP/FSDP on multi-GPU HPC clusters by tuning gradient synchronization and memory usage
- Implemented ring-attention variants in PyTorch and JAX to evaluate long-context training efficiency and scaling behavior

ML Research Assistant, Mechanistic Interpretability (LLM Post-Training) Jan 2025 – May 2025
Rochester Institute of Technology

- Built a profiling framework comparing GPT-2 behavior before and after fine-tuning using attention and activation-level analysis
- Identified task-specific summarization circuits in layers 2, 3, and 5; 62% of attention heads showed focused behavior post fine-tuning
- Applied targeted LoRA to Q/V projections, achieving 40% faster convergence (6 vs. 10 epochs) with 75% fewer trainable parameters
- Built reproducible benchmarking scripts to evaluate convergence, quality, and performance trade-offs across training configurations

ML Research Assistant, LLM Evaluation & Robustness Oct 2024 – Aug 2025
Rochester Institute of Technology, Office of the Provost

- Built automated evaluation harnesses for GPT-4, Claude, and Llama across reasoning, factuality, and tool-calling reliability tasks
- Created workload analysis scripts comparing backend latency and quality trade-offs to prioritize optimization opportunities
- Developed retrieval-augmented prompting strategies that reduced hallucination by 35% on factual QA benchmarks

Natural Language Processing Intern Nov 2021 – Feb 2022
Textify AI, Remote

- Fine-tuned transformer models for production NLP pipelines and deployed inference services with sub-100ms latency
- Built backend API components for model integration and monitoring to improve reliability of releases

Machine Learning Intern Jun 2021 – Aug 2021
Defence Research and Development Organisation (DRDO), New Delhi, India

- Developed probabilistic sensor-fusion models and state-estimation pipelines, improving inference speed by 30%
- Refactored ML prototypes for edge deployment, reducing deployment overhead by 40%

PROJECTS & PUBLICATIONS

Mechanistic Interpretability of GPT-like Models – arXiv:2505.17073 2025
First-author work on transformer training dynamics and targeted LoRA interventions for efficient post-training

Advancing Explainability in Neural Machine Translation – arXiv:2412.18669 2024
Introduced attention entropy and alignment agreement metrics for transformer interpretability on WMT14

Automatic Short Answer Grading System – IEEE AIC 2023 2023
Built LSTM-based sequence modeling system for automated short-answer grading